



Research article

Evaluating diabetes risk: Bayesian hierarchical models and machine learning integration

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ABSTRACT

Type 2 diabetes mellitus (T2DM) is a global health concern influenced by factors such as obesity, sedentary behavior, and poor diet. This study aims to analyze both regional and individual predictors of T2DM in Bangladesh by integrating Bayesian hierarchical models with machine learning methods. This study uses data from the 2017-18 Bangladesh Demographic and Health Survey (BDHS). Employing a Bayesian multinomial mixed-effects model, we account for regional variability and individual factors like age, gender, BMI, residence, wealth, education, employment, and hypertension. Our results indicate significant regional differences and associations between demographic and health-related factors with diabetes risk. Younger individuals and those with a higher BMI are more likely to develop diabetes, and hypertension significantly increases the risk. We applied machine learning (ML) models, including logistic regression, decision trees, k-nearest neighbor, linear discriminant analysis, and random forest, to classify diabetic status using these predictors, assessing their accuracy through 10-fold cross-validation. Logistic regression and linear discriminant analysis demonstrated robust performance across various response distributions. Simulation studies further examined the impact of different response distributions on model performance, revealing significant differences in classification accuracies. This approach of estimating parameters with a Bayesian model, applying ML for prediction, and conducting simulation studies to explore various scenarios highlights the importance of integrating these methodologies for effective diabetes prediction, providing insights for public health strategies to mitigate T2DM's impact.

1. Introduction

Type 2 diabetes mellitus (T2DM) is a chronic metabolic disorder marked by insulin resistance and relative insulin deficiency. It has emerged as a global epidemic, significantly affecting public health and economic systems worldwide. The prevalence of T2DM has been rising steadily, with factors such as obesity, sedentary lifestyles, and poor dietary habits being major contributors. These factors contribute to metabolic disorders and insulin resistance, which are key elements in the development of T2DM. A study highlights the relationship between lifestyle factors and diabetes, emphasizing the importance of managing body weight and increasing physical activity to mitigate the risk of T2DM [1]. T2DM is associated with severe complications including cardiovascular diseases, neuropathy, nephropathy, and retinopathy, leading to increased morbidity and mortality rates [2,3]. The economic burden of T2DM is substantial due to direct medical costs and indirect costs related to loss of productivity and disability [4,5]. Effective management strategies,

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including lifestyle interventions, pharmacotherapy, and regular monitoring, are crucial to mitigate its impact. Recent advances in medical research emphasize the importance of early detection and personalized treatment approaches to improve patient outcomes and reduce the healthcare burden associated with T2DM [6].

Recent advancements in diabetes research and management have increasingly focused on the application of machine learning (ML) and artificial intelligence to improve predictive accuracy and patient outcomes. Several studies have explored various methodologies and their impacts, providing a comprehensive understanding of the potential benefits and limitations of these technologies. Islam et al. developed a Clinical Decision Support System (CDSS) aimed at predicting T2DM using ML algorithms [7]. They employed algorithms such as decision trees (DT), random forest (RF), and support vector machines (SVM) to enhance predictive accuracy. Their findings underscore the potential of ML in early diabetes detection, thereby improving patient outcomes through timely intervention and personalized treatment plans. Conversely, Tasin et al. focused on the application of ML and explainable AI for diabetes prediction, emphasizing the importance of interpretability in AI models to ensure that healthcare professionals can understand and trust the predictions [8]. By utilizing algorithms like logistic regression (LR), DT, and gradient boosting machines (GBM), they developed robust predictive models, balancing accuracy and explainability essential for practical deployment in clinical settings.

The prevalence of T2DM has been rapidly increasing over the past few decades in Bangladesh. The lifestyle of the population, including urbanization, dietary shifts, and reduced physical activity, has contributed to this rise [9]. A recent systematic review reveals that approximately 7.1 million people in Bangladesh are suffering from this disease [10]. Other studies have indicated that the prevalence of T2DM in Bangladesh has reached concerning levels, affecting both urban and rural populations [11].

Numerous studies have been conducted on Bangladesh Demographic and Health Survey (BDHS) dataset [12,13] and Pima Indian diabetes dataset [7,8,14] to predict diabetes in Bangladesh using ensemble and ML approaches. A recent study indicated a significant variation in T2DM prevalence among different divisions in Bangladesh, using a quantile regression approach to identify risk factors for it [13]. Islam et al. used the Bagged CART algorithm to analyze and predict diabetes with the help of the Pima Indian Diabetes Dataset [12]. This study employed SVM, RF, LR, linear discriminant analysis (LDA), k-nearest neighbor (KNN), bagged classification, and regression tree (Bagged CART) to predict diabetes in the ML framework, achieving the highest accuracy of 94.3 % from the Bagged CART model. However, the dataset used in their study had an imbalanced distribution of the response variable, with approximately 92 % of the cases being non-diabetic and only 8 % diabetic. This imbalance could affect the performance of the ML models by making it harder to accurately predict the smaller diabetic class.

In this paper, we employed a Bayesian multinomial mixed-effects model to capture both fixed and random effects, accounting for the hierarchical structure of the data and regional variations. This approach allows for a more nuanced understanding of regional variability by treating it as a random effect, which contrasts with the fixed effects commonly used in previous literature. To obtain reliable and accurate parameter estimates, we utilized weakly informative priors, which provide a balance between data-driven insights and prior knowledge without imposing overly restrictive assumptions. To classify and predict diabetic status, we used several ML models, including LR, DT, KNN, LDA, and RF. To ensure that our models were both robust and accurate, we performed extensive simulation studies. These simulations involved testing the models on various distributions of the response variable, allowing us to evaluate how well each model performed under different conditions. These simulations also revealed how prediction accuracy varies across different models, highlighting the strengths and limitations of each approach under varying conditions. This integrated methodology underscores the importance of combining Bayesian and ML techniques to enhance diabetes prediction and provides valuable insights for developing effective public health strategies.

2. Materials and methods

2.1. Study design

In this study, we considered the data from the 2017-18 BDHS dataset, carried out in collaboration with the Ministry of Health and Welfare, Mitra and Associates, NIPORT, and USAID [15]. This nationwide survey utilized a two-stage stratified sampling design. Initially, 672 primary sampling units (PSUs) were selected, comprising 423 clusters from rural areas and 249 from urban areas. In the second stage, 30 households were systematically chosen from each PSU, leading to a total of 4864 households eligible for biomarker measurements from the initially selected 20,160 families. The study included 14,704 individuals (6691 men and 8013 women) aged 18 years or older. Due to the incompleteness of the responses, we selected a dataset containing information on 12,278 individual members (83.5 %) that are clustered within the Divisions (Dhaka, Khulna, Chattogram, Rangpur, Sylhet, Barisal, Mymensingh, Rajshahi) of Bangladesh. The rationale of excluding the incomplete cases is to ensure that the analysis is based on high-quality data, which can lead to more reliable and valid results. Incomplete responses can introduce noise and uncertainty, potentially compromising the integrity of the findings. Modeling with complete cases is a common practice in epidemiological and health research when dealing with large datasets. This approach is consistent with previous studies that aim to maintain the quality and reliability of the data analysis. Schafer and Graham indicated that complete case analysis remains a standard practice in many fields, provided the assumptions about missingness are carefully considered [16]. The entire analysis has been conducted using the R programming language. The variable selection for the independent variables was carried out based on their availability in the survey. In the list of predictor variables, demographics such as age, sex, educational status, and residential status are selected. We also considered socioeconomic status (economic and employment status), and health status (BMI and hypertension) of individual household members. The response variable Diabetic status is trinary that indicates whether an individual is classified as diabetic or not as well as prediabetic, based on their plasma blood glucose level. If the plasma blood glucose level is 5.6–6 mmol/dl, the individual is labeled "non-diabetic" and if the plasma blood glucose level is 6.1–6.9 mmol/dl then the individual is labeled as "prediabetic", again if the plasma blood glucose level is

greater than 7 mmol/dl, the individual is labeled "diabetic" [17]. The predictor variables and categories are Divisions (Dhaka, Khulna, Chattogram, Rangpur, Sylhet, Barisal, Mymensingh, Rajshahi), Residence (Urban, Rural), Wealth (Richest, Rich, Middle, Poor, Poorest), Gender (Male, Female), Age (24 or Less: Adult group, 25–44: Young group, 45–64: Middle Age group, 65 or More: Old group [18]), Education (No Education, Primary, Secondary, Higher), Employment (Yes: who has formal job, No: otherwise), BMI (Underweight: less than 18.5 kg/m², Normal: 18.5–24.9 kg/m², Overweight: 25–29 kg/m², Obese: more than 30 kg/m²), Hypertension (Yes: systolic blood pressure ≥ 140 mmHg or diastolic blood pressure ≥ 90 mmHg, No: otherwise).

Table 1 provides the univariate distribution of demographic, socioeconomic, and health profile of 12,278 individuals from the 2017–18 BDHS. The data are distributed across eight divisions: Barisal (10.4 %), Chittagong (13.5 %), Dhaka (13.6 %), Khulna (13.6 %), Mymensingh (11.5 %), Rajshahi (13 %), Rangpur (12.6 %), and Sylhet (11.9 %). The population is predominantly rural (63.8 %) compared to urban (36.2 %). Socioeconomic status, as indicated by wealth quintiles, shows a relatively even distribution, with the richest (22.5 %) slightly outnumbering other categories. Gender distribution reveals more males (56.5 %) than females (43.5 %). Age distribution highlights that the majority of the population falls into the middle age group (44.5 %), followed by old people (25.4 %), young (20.5 %), and adults (9.6 %). Educational attainment varies, with secondary education (30.3 %) being the most common, followed by higher education (28.7 %), no education (24.5 %), and primary education (16.5 %). Employment status shows that a significant portion of the population is not working (60.7 %). Health indicators reveal that the majority have normal BMI (59.3 %), with fewer individuals classified as underweight (4.4 %), overweight (18.8 %), and obese (17.4 %). A significant proportion of the population suffers from hypertension (75.8 %). Diabetic status indicates that most individuals are non-diabetic (75.2 %), while diabetic and prediabetic statuses are equally distributed (12.4 % each). The proportion test p-values (<0.001) for each variable confirm that the distributions across these categories are significantly unequal.

Table 2 reveals the bivariate relationship of the predictors among different levels of the response variable. The detailed examination of the dataset provides insightful revelations into the distribution of diabetic and prediabetic statuses across various demographic, socioeconomic, and health-related factors. Within the divisions, Dhaka exhibits the highest proportion of diabetic individuals (21.7 %) and prediabetic individuals (19.7 %), indicating a significant regional disparity when compared to other divisions like Rangpur, which has the lowest proportion of diabetic (6.5 %) and prediabetic (8.7 %) individuals. This result motivates to consider the division as cluster in further analysis. The residence data highlights a notable difference between urban and rural areas, with urban residents showing a higher prevalence of diabetes (16.3 %) compared to their rural counterparts (10.3 %). Socioeconomic status, as

Table 1
Distribution of the variables in the dataset.

Variables	Categories	Frequency	Percentages	Proportion Test (p value)
Division	Barisal	1272	10.4	<0.001
	Chittagong	1661	13.5	
	Dhaka	1675	13.6	
	Khulna	1665	13.6	
	Mymensingh	1408	11.5	
	Rajshahi	1591	13	
	Rangpur	1545	12.6	
	Sylhet	1461	11.9	
Residence	Rural	7837	63.8	<0.001
	Urban	4441	36.2	
Wealth index	Poorest	2419	19.7	<0.001
	Middle	2315	18.9	
	Poorer	2389	19.5	
	Richest	2389	19.5	
	Richer	2766	22.5	
Gender	Male	6943	56.5	<0.001
	Female	5335	43.5	
Age	Adult	1174	9.6	<0.001
	Young	2518	20.5	
	Middle	5465	44.5	
	Old	3121	25.4	
Education	Primary	2025	16.5	<0.001
	No Education	3009	24.5	
	Secondary	3715	30.3	
	Higher	3529	28.7	
Employment	No	7458	60.7	<0.001
	Yes	4820	39.3	
BMI	Normal Weight	7285	59.3	<0.001
	Underweight	537	4.4	
	Overweight	2314	18.8	
	Obese	2142	17.4	
Hypertension	No	2971	24.2	<0.001
	Yes	9307	75.8	
Diabetic Status	Diabetic	1527	12.4	<0.001
	Normal	9232	75.2	
	Prediabetic	1519	12.4	

Table 2

Distribution of the variables among different levels of the response variable.

Variables	Categories	Diabetic N (%)	Non-Diabetic N (%)	Prediabetic N (%)	Chi Square (p value)
Division	Barisal	152(11.9 %)	925(72.7 %)	195 (15.3 %)	<0.001
	Chittagong	237(14.3 %)	1199(72.2 %)	225(13.5 %)	
	Dhaka	363(21.7 %)	982(58.6 %)	330 (19.7 %)	
	Khulna	180(10.8 %)	1322(79.4 %)	163(9.8 %)	
	Mymensingh	158(11.2 %)	1093(77.6 %)	157(11.2 %)	
	Rajshahi	159(10.0 %)	1288(81.0 %)	144(9.1 %)	
	Rangpur	100(6.5 %)	1311(84.9 %)	134(8.7 %)	
	Sylhet	178(12.2 %)	1112(76.1 %)	171(11.7 %)	
Residence	Rural	805(10.3 %)	6105(77.9 %)	927(11.8 %)	<0.001
	Urban	722(16.3 %)	3127(70.4 %)	592(13.3 %)	
Wealth index	Poorest	178(7.5 %)	1949(81.6 %)	262(11.0 %)	<0.001
	Middle	244(10.1 %)	1923(79.5 %)	252(10.4 %)	
	Poorer	186(8.0 %)	1904(82.2 %)	225(9.7 %)	
	Richest	606(21.9 %)	1701(61.5 %)	459(16.6 %)	
Gender	Richer	313(13.1 %)	1755(73.5 %)	321(13.4 %)	<0.001
	Male	671(12.6 %)	4010(75.2 %)	654(12.3 %)	
Age	Female	856(12.3 %)	5222(75.2 %)	865(12.5 %)	<0.001
	Adult	207(8.2 %)	2070(82.2 %)	241(9.6 %)	
Education	Young	627(11.5 %)	4150(75.9 %)	688(12.6 %)	<0.001
	Middle	519(16.6 %)	2163(69.3 %)	439(14.1 %)	
	Old	174(14.8 %)	849(72.3 %)	151(12.9 %)	
	Primary	459(12.4 %)	2816(75.8 %)	440(11.8 %)	
	No Education	351(11.7 %)	2270(75.4 %)	388(12.9 %)	
Employment	Secondary	444(12.6 %)	2653(75.2 %)	432(12.2 %)	<0.001
	Higher	273(13.5 %)	1493(73.7 %)	259(12.8 %)	
	Work	719(14.9 %)	3477(72.1 %)	624(12.9 %)	
	No Work	808(10.8 %)	5755(77.2 %)	895(12.0 %)	
BMI	Normal Weight	818(11.2 %)	5617(77.1 %)	850(11.7 %)	<0.001
	Underweight	175(8.2 %)	1734(81.0 %)	233(10.9 %)	
	Overweight	413(17.8 %)	1566(67.7 %)	335(14.5 %)	
	Obese	121(22.5 %)	315(58.7 %)	101(18.8 %)	
Hypertension	No	1000(10.7 %)	7176(77.1 %)	1131(12.2 %)	<0.001
	Yes	527(17.7 %)	2056(69.2 %)	388(13.1 %)	

reflected by wealth quintiles, reveals that the richest individuals have the highest rates of diabetes (21.9 %) and prediabetes (16.6 %), while the poorest groups exhibit the lowest diabetic rates (7.5 %). Gender distribution shows similar rates of diabetes and prediabetes between males and females. Age-wise, the middle age group has the highest prevalence of diabetes (16.6 %), while the adult group has the lowest (8.2 %). Educational attainment presents an interesting pattern where individuals with higher education levels have slightly higher diabetic rates (13.5 %) compared to those with no education (11.7 %). Employment status indicates that those not working have a higher prevalence of diabetes (14.9 %) compared to those who are employed (10.8 %). BMI and hypertension status are significant predictors, with obese individuals and those with hypertension showing markedly higher rates of diabetes (22.5 % and 17.7 %, respectively) compared to their normal-weight and non-hypertensive counterparts. These findings are statistically significant with p-values <0.001 across all variables, underscoring the critical associations between these factors and the diabetic and prediabetic statuses in the Bangladeshi population.

2.2. Bayesian multinomial mixed effect model

We applied Bayesian multinomial mixed-effects model considering the division as random effect and the other predictors as fixed effect to identify their effect (odds ratio) on the diabetic status. This is crucial in capturing the intrinsic variability within and between clusters, leading to more accurate and robust inference. Bayesian methods offer significant advantages in this context, including the ability to incorporate prior knowledge into the analysis and to provide full probabilistic interpretations of model parameters. These methods are increasingly recommended in epidemiological and public health research for their flexibility and comprehensiveness [19, 20]. The multinomial logistic component of the model is appropriate for the trinary response variable (diabetic, prediabetic, and non-diabetic) in our dataset. This allows us to simultaneously model the probabilities of multiple outcome categories, addressing the complexity of diabetes status classification in the population. The mixed-effects component, on the other hand, accounts for the hierarchical structure of the data, such as individuals nested within divisions, thus capturing both within-division and between-division variability effectively. For the predictor variables, we considered weakly informative prior from the literature [21]. For the diabetic response, we used prior distributions for BMI [22], age, gender, and employment status [23] from the literature, where the researchers elucidate the differential impacts of these factors on diabetes prevalence. For socioeconomic factors such as residence and wealth, we used findings from a study by Hossain et al. (2019) that highlights the association between these variables and diabetes risk in a population-based study in Bangladesh [24]. Educational status priors were guided by the work of Reiner et al. (2019), which examines

the influence of education on health outcomes, including diabetes [25]. Finally, hypertension, a significant comorbidity with diabetes, was informed by the study conducted by Whelton et al. (2017), which provides detailed insights into the relationship between hypertension and diabetes [26]. For the prediabetic response, we employed prior distributions from studies that specifically examine the predictors of prediabetes. Age, residence, employment, and wealth were informed by a study on the socioeconomic and demographic determinants of prediabetes by Shaikh et al. (2023), which provides detailed statistical associations for these factors [27]. For BMI, education, gender, and hypertension, we referred to a study by Akter et al. (2021), which comprehensively examines these variables in the context of prediabetes and provides robust prior estimates [10].

The use of Bayesian multinomial mixed-effects models with weakly informative priors helps us to mitigate the impact of selection bias that has been arise due to the exclusion of missing responses. Gelman et al. highlighted that Bayesian approaches can improve model estimates and predictions, particularly in the presence of complex data structures and potential biases [19]. Because the models can incorporate prior knowledge and use hierarchical structures to account for unobserved heterogeneity, leading to robust parameter estimates even with incomplete data. Gelman and Hill also emphasized the importance of diagnostic checks in confirming the validity of Bayesian hierarchical models [28]. These checks validate that the model captures the underlying data structure accurately. We conducted diagnostic checks, such as prior predictive checks, posterior predictive checks and examining model convergence, to ensure that the model performs well despite the exclusion of incomplete cases. To assess the robustness of this model, we carried out sensitivity analyses under both regularising and diffuse prior specifications. The resulting estimates and their associated prior predictive checks are provided in [Table S1](#) and [Fig. S3](#) of the Supplementary Materials, while the full computational procedure of this model is delineated in Algorithm S2.

2.3. Classification using machine learning model

Considering the need for precise classification of diabetic status, the application of ML models offers a promising approach to enhance predictive accuracy and model robustness. ML techniques have been increasingly recognized for their ability to handle complex, non-linear relationships among variables, which is particularly beneficial in the context of predicting health outcomes like diabetes and prediabetes [29]. These models can process large datasets with numerous predictors, capturing subtle interactions that traditional statistical methods might overlook [30]. For instance, LR, while being a robust method for binary classification, can be extended to multinomial LR to handle our trinary outcome of diabetic status [31]. DT and RF, which are based on the principle of recursive partitioning, provide intuitive models that can easily accommodate interaction terms and non-linear effects [32]. On the other side, KNN and LDA are also valuable for their simplicity and effectiveness in classification tasks [33]. KNN is particularly useful when the decision boundary is irregular, while LDA performs well when the predictor variables are approximately normally distributed within each class [34]. The integration of these ML models not only improves the classification performance but also allows for more flexible handling of diverse data types and distributions.

Given the complex interplay of demographic, socioeconomic, and health-related factors in predicting diabetic status, we applied these ML models to provide a more nuanced understanding and better predictive performance compared to traditional methods. We considered 80 % data on training set and 20 % on test set and also did 10-fold cross validation in each case to identify the test accuracy. This rationale aligns with contemporary research emphasizing the utility of ML in public health and epidemiology, as highlighted in studies like those by Beam and Kohane [35]. In this study, we explored two distinct scenarios for the response variable: binomial and trinomial classifications. In the trinomial scenario, we included all three categories of the response variable: non-diabetic, prediabetic, and diabetic. Conversely, in the binomial scenario, we combined the non-diabetic and prediabetic groups into a single category. The rationale for considering these two scenarios is that this approach stems from the distribution of data within the categories: diabetic, prediabetic, and non-diabetic. Given that the percentages of data (in [Table 1](#)) in the diabetic and prediabetic categories are similar and smaller compared to the non-diabetic category, merging the diabetic and prediabetic groups to form a binomial response allows us to explore whether this aggregation improves the prediction accuracy of the models. This is particularly important because the imbalanced nature of the dataset, with a dominant non-diabetic category, can pose challenges for classification algorithms, potentially leading to biased or less accurate predictions. By evaluating the models using both binomial and trinomial responses, we can assess how well they handle class imbalance and whether combining categories provides a more stable and accurate prediction framework. This dual approach allows us to identify the optimal model configuration that balances accuracy and robustness in predicting diabetic status. Additionally, this methodology aligns with practices in medical research where different classification schemes are tested to ensure comprehensive evaluation of predictive performance. This approach has been supported in literature, where researchers have demonstrated that handling class imbalance and testing different response aggregations can significantly impact model performance [36,37].

2.4. Simulation study

To thoroughly evaluate the classification models, we simulated various scenarios for the response variable that is generated using coefficients from the Bayesian multinomial mixed-effects model. This simulation was crucial for understanding how different distributions of the response variable impact prediction accuracy. Initially, we utilized the posterior distributions of the model coefficients to generate synthetic response data. Specifically, we derived two sets of coefficients: one for the binomial response and another for the trinomial response. Using these coefficients, we created linear predictors for each category by incorporating relevant predictors such as age, gender, BMI, residence, wealth, education, employment, and hypertension. These linear predictors were then transformed using the logit function to obtain probabilities for each category. The response variable was simulated by assigning each observation to a

category based on these probabilities, ensuring predefined distributions for non-diabetic, prediabetic, and diabetic categories, respectively. This approach allowed us to generate realistic synthetic data of the response variable, reflecting the complex interplay of various predictors as modelled by the Bayesian multinomial mixed-effects model. The detailed simulation process is provided in the supplementary material (Algorithm S1).

The simulation scenarios differ only in the distribution of the data across different levels of the response variable. This methodology enabled us to robustly evaluate the performance of different ML classifiers in predicting diabetic status under varied response distributions. By doing so, our evaluation accounts for the inherent variability and distribution patterns observed in real-world data, leading to more robust and reliable model performance assessments.

3. Results

3.1. Bayesian multinomial mixed effect model result

The results of the Bayesian multinomial mixed effect model provide a detailed understanding of the variability in diabetic and prediabetic responses across different divisions of Bangladesh. The model includes random intercepts for 8 different divisions, allowing for the baseline odds of diabetes and prediabetes to vary regionally. The standard deviation (sd) of the intercept for the diabetic response is estimated at 2.81, with a 95 % credible interval (CI) ranging from 0.13 to 10.47, indicating substantial variability across divisions. Similarly, the standard deviation of the intercept for the prediabetic response is 2.70, with a 95 % CI of 0.10–10.10, again suggesting substantial regional differences. These results underscore the importance of considering regional variability when designing and implementing public health interventions aimed at managing diabetes and prediabetes. The wide credible intervals highlight the uncertainty in these estimates. This suggests that certain divisions may have higher or lower baseline odds of diabetes and prediabetes, potentially due to regional factors that were not directly measured in the study. In this study, we followed Gelman's Bayesian workflow (34) that involves an iterative cycle of model creation, inference, checking, evaluation, and refinement. In a simplified version, the process begins by defining a model informed by exploratory data analysis, often utilizing visualizations for better data comprehension. Weakly informative priors are selected to regularize the estimates without imposing excessive constraints. Prior predictive checks are then conducted to verify the reasonableness of the model's assumptions before examining the actual data. Once the priors are deemed appropriate, we proceed to estimate the posterior distribution of the model parameters through Markov Chain Monte Carlo (MCMC) simulation. Following model fitting, we perform diagnostic checks to validate the computations, which includes analyzing trace plots and ensuring convergence criteria, such as the Gelman-Rubin $\hat{R}$ statistic, are met. Next, the model is evaluated using posterior predictive checks, where the observed data is compared to data simulated from the model to identify any discrepancies. A well-fitting model will produce simulated data that closely resembles the observed data.

3.1.1. Prior predictive check

It is important in Bayesian to understand whether the specified priors are reasonable for the data and the model. Prior predictive checks help here to ensure that the priors are reasonable and do not lead to unrealistic predictions before fitting the model to the actual data. It involves generating data from the model using only the priors and comparing these simulated data to the observed data.

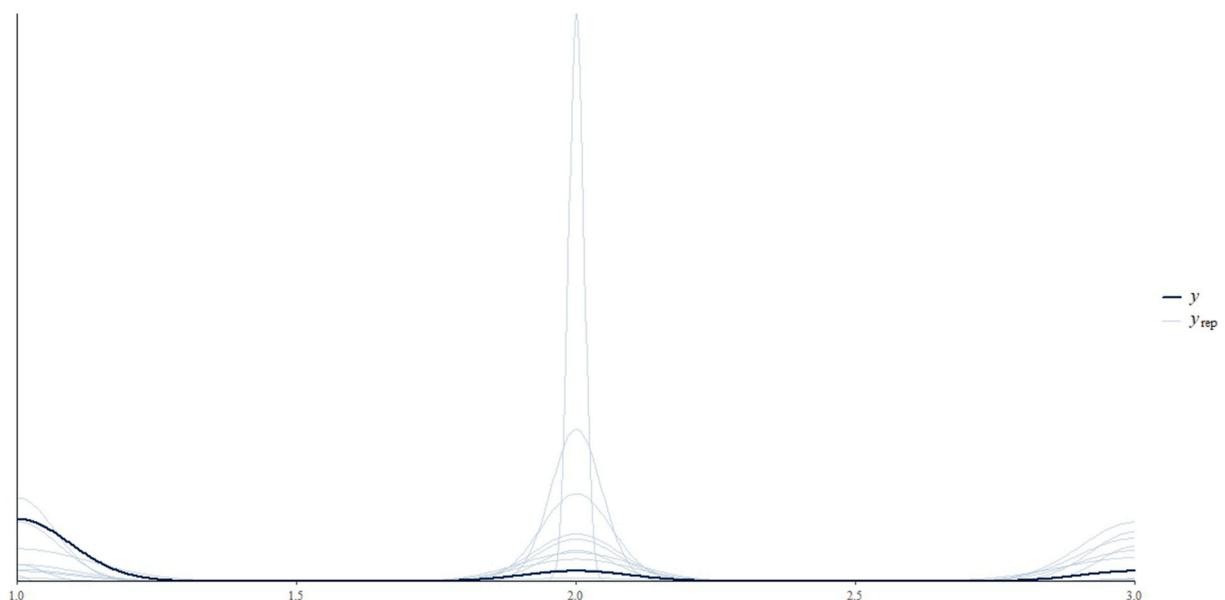


Fig. 1. Prior predictive check.

Fig. 1 presets the prior predictive checks for this study. The x axis represents the range of the predicted values, while the y axis represents the density of the predicted values. The dark blue line (y) represents the observed data, where the light blue lines (y_{rep}) represent the predicted data generated using only the prior distributions. The light blue lines broadly cover the range of the dark blue line, indicating that the priors are not too restrictive and can generate a wide range of plausible data. Ideally, the distribution of the predicted data (y_{rep}) should encompass the observed data (y) and this alignment is seen in **Fig. 1**. The peaks and valleys of the light blue lines align with those of the dark blue line, reflecting the priors' ability to model realistic patterns in the observed data. This suggests that the priors are well-calibrated and neither overly specific nor excessively vague, thereby ensuring that the model assumptions are appropriate before incorporating the actual data for parameter estimation.

3.1.2. Posterior distribution

The analysis in **Table 3**, which includes various demographic, socioeconomic, and health-related factors, reveals significant associations with both diabetic and prediabetic outcomes. For instance, gender, age, residence, wealth, education, employment, BMI, and hypertension status are all examined to understand their impact on diabetes and prediabetes. The odds ratios (OR), SE and 95 % CI offer insights into the relative likelihood of these conditions across different subgroups. We also added the figures of the posterior distribution in Supplement (Table B) to illustrate the uncertainty and variability in our estimates.

The model was applied to BDHS dataset containing 12,278 observations. In terms of fixed effects, **Table 3** provides the ORs, SE and 95 % CIs for various predictors of diabetic and prediabetic response. For gender, males have an OR of 0.97 (95 % CI: 0.9, 1.05) compared to females, indicating no significant difference in the likelihood of being diabetic. Age-wise, younger individuals (referred to as "Young") show significantly higher odds of being diabetic with an OR of 1.26 (95 % CI: 1.19, 1.34), while middle-aged individuals also have increased odds with an OR of 1.2 (95 % CI: 1.06, 1.35) compared to the adult reference group. Conversely, older individuals have lower odds of being diabetic with an OR of 0.73 (95 % CI: 0.64, 0.84). Rural residents show no significant difference from urban residents with an OR of 0.95 (95 % CI: 0.9, 1.01). Regarding wealth, all categories show higher odds compared to the poorest, with the richest having the highest OR of 1.22 (95 % CI: 1.11, 1.35). Education levels, on the other hand, do not show a consistent pattern of association with diabetes. Employment status indicates that those who are employed have a slightly higher odds of being diabetic (OR: 1.14, 95 % CI: 1.02, 1.28). In terms of BMI, overweight and obese individuals have significantly higher odds of being diabetic with ORs of 1.36 (95 % CI: 1.2, 1.51) and 1.26 (95 % CI: 1.19, 1.34), respectively. Lastly, individuals with hypertension have markedly significantly higher odds of being diabetic with an OR of 3.13 (95 % CI: 1.75, 5.64).

For the prediabetic response, the model indicates that younger individuals have significantly higher odds of being prediabetic, with an OR of 1.95 and a 95 % CI of 1.48–2.59. Middle-aged individuals also have significantly higher odds with an OR of 1.95 (95 % CI: 1.49, 2.56). Rural residence is associated with increased odds of prediabetes compared to urban areas, with an OR of 1.22 (95 % CI: 0.87, 1.72). Wealth categories except poorer group show increased odds of being prediabetic compared to the poorest group, with the highest wealth quintile having an OR of 1.79 (95 % CI: 1.13, 2.80). For BMI, obese individuals have higher odds of being prediabetic

Table 3
Result of Bayesian multinomial mixed effect models.

Predictor	Levels	Diabetic Response			Prediabetic Response		
		OR	SE	95 % CI	OR	SE	95 % CI
Intercept		0.65 (Ref ^a)	48.91	(0.00, 1702.75)	0.35	42.1	(0.00, 544.57)
Gender	Female						
	Male	0.97	1.04	(0.90, 1.05)	0.98	1.12	(0.79, 1.22)
Age	Adult						
	Young	1.26	1.03	(1.19, 1.34)	1.95	1.15	(1.48, 2.59)
	Middle	1.20	1.06	(1.06, 1.35)	1.95	1.15	(1.49, 2.56)
	Old	0.73	1.07	(0.64, 0.84)	1.95	1.31	(1.15, 3.39)
Residence	Urban						
	Rural	0.95 (Ref ^a)	1.03	(0.90, 1.01)	1.22	1.19	(0.87, 1.72)
Wealth index	Poor						
	Poorer	1.09 (Ref ^a)	1.05	(0.99, 1.21)	0.94	1.23	(0.63, 1.43)
	Middle	1.13	1.06	(1.00, 1.26)	1.22	1.23	(0.81, 1.86)
	Richer	1.17	1.05	(1.06, 1.30)	1.38	1.23	(0.92, 2.08)
	Richest	1.22	1.05	(1.11, 1.35)	1.79	1.27	(1.13, 2.80)
Education	No						
	Primary	0.57 (Ref ^a)	1.34	(0.32, 1.01)	1.20	1.21	(0.83, 1.73)
	Secondary	0.73	1.35	(0.40, 1.34)	1.06	1.16	(0.79, 1.42)
	Higher	0.67	1.35	(0.38, 1.20)	1.06	1.17	(0.77, 1.45)
Employment	No						
	Yes	1.14 (Ref ^a)	1.06	(1.02, 1.28)	0.95	1.13	(0.75, 1.21)
BMI	Normal						
	Underweight	1.30 (Ref ^a)	1.14	(1.00, 1.67)	0.97	1.15	(0.74, 1.27)
	Overweight	1.36	1.05	(1.25, 1.51)	1.08	1.12	(0.87, 1.34)
	Obese	1.26	1.03	(1.19, 1.34)	1.31	1.19	(0.95, 1.84)
Hypertension	No						
	Yes	3.13 (Ref ^a)	1.35	(1.75, 5.64)	0.98	1.12	(0.79, 1.21)

^a Reference Category.

(OR: 1.31, 95 % CI: 0.95, 1.84). The results also indicate that hypertension is not significantly associated with prediabetes, with an OR of 0.98 (95 % CI: 0.72, 1.21).

3.1.3. Model diagnostics checks

The histograms for the predictors in Supplement (Table A) provide a comprehensive overview of the posterior distributions and trace plots for various predictors in the Bayesian multinomial mixed-effects model, which includes diabetic, prediabetic, and non-diabetic categories. The histograms on the left show the posterior distributions for the coefficients of the model, while the trace plots on the right depict the convergence of the MCMC chains. The posterior distributions reveal the estimated effect sizes for each predictor, with the distribution's spread indicating the uncertainty around these estimates. For instance, the posterior distributions for the intercepts of both diabetic and prediabetic responses appear centered around zero, suggesting no substantial baseline difference in odds. The trace plots for these intercepts, as well as other predictors, demonstrate that the MCMC chains have mixed well and converged, as evidenced by their consistent patterns across iterations and between chains. Predictors like residence, wealth, gender, age, education, employment, BMI, and hypertension status exhibit varying degrees of influence on diabetic and prediabetic outcomes. Notably, the posterior distributions for BMI categories such as obese and overweight show higher values, indicating a strong positive association with diabetic status. Similarly, the wealth categories like richest show significant positive effects, suggesting higher odds of diabetes in wealthier individuals. In contrast, predictors like education (higher) show negative associations with diabetic status. For random effects, the standard deviations for the intercepts across different divisions are also shown, indicating variability in baseline diabetic and prediabetic probabilities across these regions. The histograms for these random effects suggest a wide range of variation, while the trace plots confirm that the MCMC chains for these random effects have converged satisfactorily. The variability observed here seems to be consistent with typical MCMC behavior when exploring a posterior with a certain degree of uncertainty. The coefficient for all the predictors displays tight clustering around a particular value, demonstrating precision in the posterior estimate. The chain exhibits no apparent trends, supporting the conclusion that it has reached a stable distribution.

The metrics $\hat{R}$ (potential scale reduction factor) and Effective Sample Size (ESS) are also crucial diagnostics for MCMC convergence. $\hat{R}$ values close to 1 (typically ≤ 1.1) indicate convergence, while ESS quantifies the number of independent-like draws from the posterior distribution. Both metrics can often be visualized as part of summary tables or alongside other plots to provide a quick diagnostic overview.

Fig. 2 appears to be a visualization of the $\hat{R}$ statistic, which is used to assess the convergence of the MCMC chains in our Bayesian model. The $\hat{R}$ statistic compares the variation within chains to the variation between chains, with values close to 1 (typically ≤ 1.05) indicating good convergence. In this figure, all $\hat{R}$ values are below the 1.05 threshold, confirming that the chains have mixed well and are sampling from the same posterior distribution. There are no points beyond this threshold, which is a strong indication that the MCMC algorithm has successfully captured stable and reliable parameter estimates. In plain terms, this means that the model has reached a state where its estimates are consistent and trustworthy, ensuring that the results drawn from the posterior distributions are robust and can be confidently interpreted.

Fig. 3 represents the Effective Sample Size (ESS) for the MCMC chains in our model, normalized by the total number of post-warmup iterations (N_{eff}/N). ESS measures how many independent-like samples are effectively obtained from the posterior

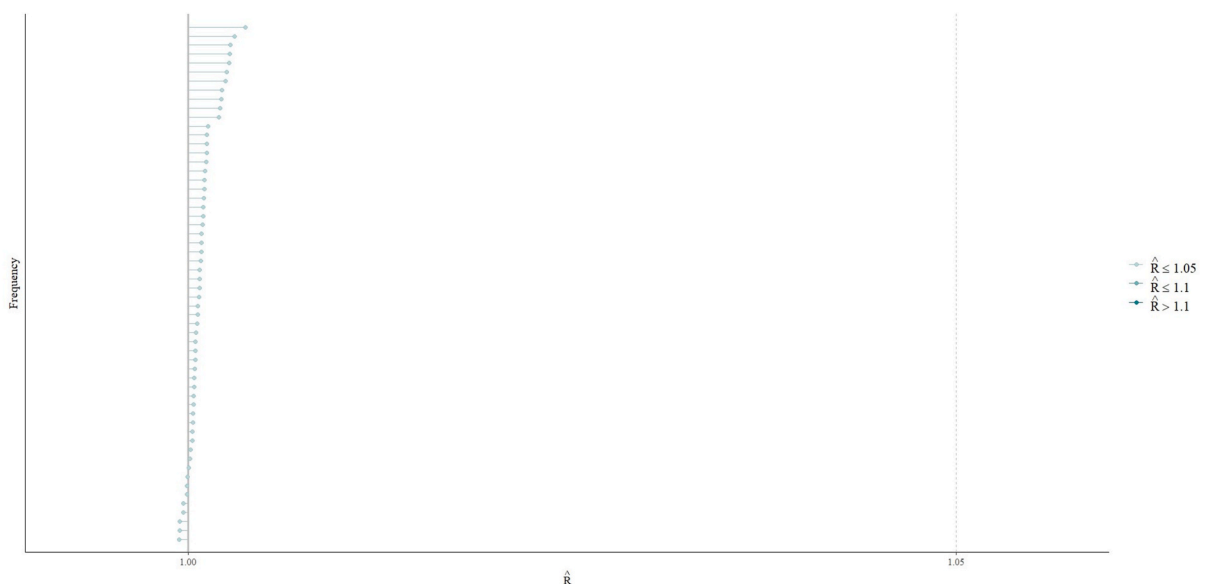


Fig. 2. Gelman–Rubin statistic ($\hat{R}$) across parameters.

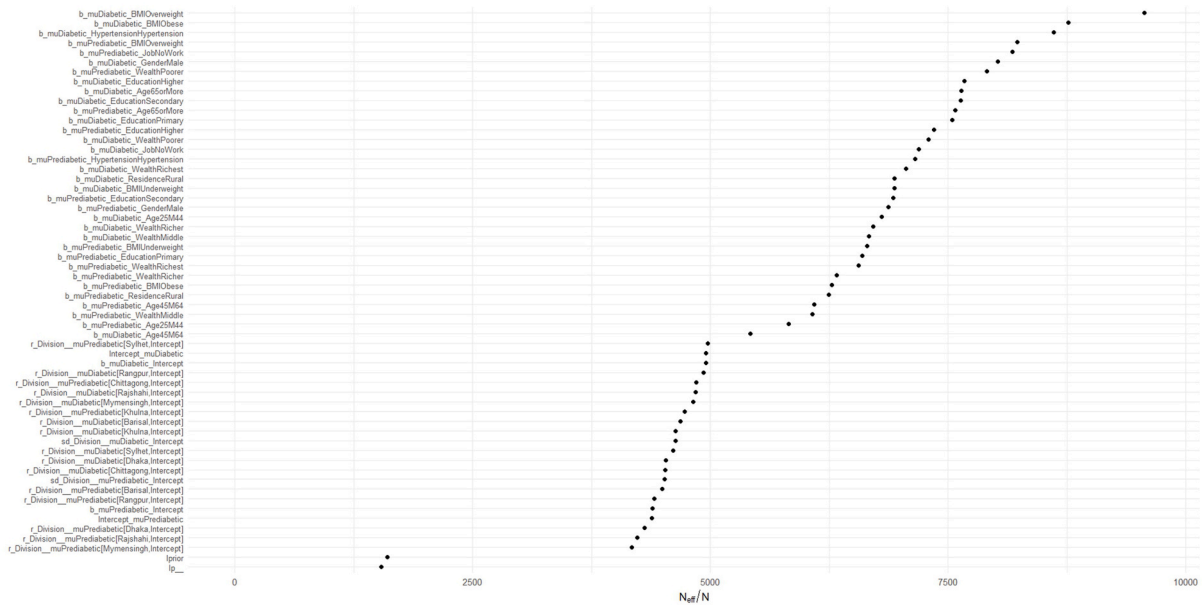


Fig. 3. ESS check for model convergence evaluation.

distribution, even though MCMC chains produce correlated samples. The plot visualizes the ESS ratios $\left(N_{\text{eff}}/N\right)$ for various model parameters. On the x-axis, we have the effective sample size ratios, and on the y-axis, the model parameters are listed. This plot reveals that the majority of the parameters have a high ESS ratio, indicating that the MCMC chains have mixed well and provide reliable estimates. Parameters such as the intercepts and division-specific random effects have particularly high ESS values, suggesting that these estimates are very robust. However, a few parameters with lower ESS values suggest that additional iterations could improve the reliability of those estimates. In simpler terms, this figure demonstrates how efficiently the MCMC process has generated enough high-quality samples for the model to produce accurate results. High ESS values mean that we can confidently trust the model's inferences, while lower values act as a reminder to double-check those parameters or run the simulation longer for additional robustness.

Posterior predictive checks are powerful tools for validating Bayesian models that ensure the inferences drawn from the model are robust and reflective of the observed data. Posterior predictive checks are motivated by the need to verify that the model fits the data well. The central idea is to compare the observed data to data simulated from the model's posterior predictive distribution. If the model

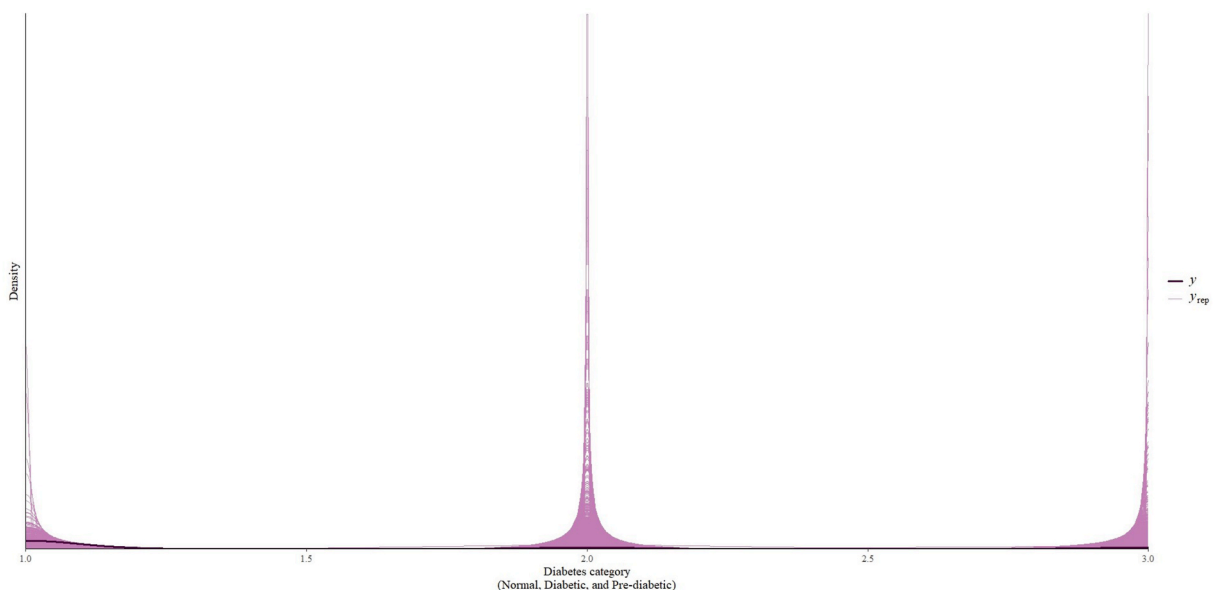


Fig. 4. Posterior predictive checks for model performance evaluation.

is a good fit, the simulated data should look like the observed data across various aspects. In particular, posterior predictive checks allow us to assess whether the model's assumptions are justified and whether the model captures the data structure effectively. By visualizing discrepancies between observed and simulated data, posterior predictive checks can highlight specific areas where the model may be lacking.

Fig. 4 (Posterior predictive checks of our model, which are used to evaluate how well the model fits the observed data) shows both the observed data (likely represented by the darker line labeled y) and a set of simulated data sets from the posterior predictive distribution (the lighter lines labeled y_{rep}). The key idea is that if the model is a good fit, the simulated data (y_{rep}) should closely resemble the observed data (y) in terms of patterns and spread. In this case, the lines (y_{rep}) largely overlap with the (y) line, indicating that the model is accurately capturing the structure of the observed data. The clustering of the (y_{rep}) lines around the (y) line reflects low uncertainty in the model's predictions, meaning that the model is making consistent and reliable predictions based on the observed data. If the (y_{rep}) lines were scattered far from the (y) line, it would suggest that the model is failing to capture the key characteristics of the observed data. This alignment of the observed and simulated data confirms that the model's assumptions and structure are appropriate for the dataset, providing confidence in the robustness of the model's results.

3.1.4. Sensitivity to prior choice

We also checked the sensitivity of the Bayesian multinomial mixed effect models by using different set of priors. Prior predictive checks (PPC) across alternative priors (Fig. S3 in the supplement) further confirm the robustness of the model. PPC with informative prior was generated from the model discussed in Table 3. PPC with regularising prior replaces those prior means with 0 and shrinks the standard deviations by 50 %, representing a deliberately sceptical stance; PPC with wider prior keeps the means at 0 but doubles the standard deviations, allowing substantially larger effect sizes. Across all three settings the 90 % prior-predictive intervals (light bars) encompass the observed class counts (dark points) for each of the eight divisions, and the posterior means align closely regardless of prior width. No region shows a systematic shift in the relative frequencies of normal, pre-diabetic or diabetic categories when the priors are tightened or relaxed. Table 3 reports the coefficients from the informative prior model. When those same predictors are re-estimated with regularising and diffuse priors (Table S1 in the supplement), posterior medians shift by ≤ 0.05 on the log-odds scale and every 95 % credible interval continues to encompass Table 3. Direction and magnitude of the key effects—age, BMI category and hypertension—remain unchanged. On the other hand, LOO cross validation [38] supports the same conclusion that the expected log-predictive density (ELPD) for the Informative-prior model is only 0.1 higher than that of the Regularising model and 0.3 higher than the Diffuse model, with corresponding standard errors of 0.1. Because each difference is smaller than its own uncertainty, none of the models can be said to predict meaningfully better than another. Thus, predictive accuracy remains essentially identical across the three prior specifications, reinforcing that our substantive conclusions are not driven by prior choice. Multi-class Brier scores told the same story too: 0.128, 0.129 and 0.13 for the Informative, Regularising and Diffuse prior models, respectively—differences well below 0.01. Predictive calibration therefore remains virtually unchanged as prior informativeness varies.

3.2. Classification of diabetic status

The results of the 10-fold cross-validation for test accuracy across different ML models using BDHS data are summarized in Table 4. The models evaluated include LR, DT, KNN, LDA, and RF, each tested on both trinomial (model 1) and binomial (model 2) classification tasks. For trinomial classification, KNN slightly outperformed other models with an accuracy of 0.753, while LR, DT, and LDA each achieved an accuracy of 0.752. The RF model had a slightly lower accuracy at 0.743. In the binomial classification task, LR, DT, KNN, and LDA all achieved an accuracy of 0.875, whereas RF had a marginally lower accuracy of 0.869.

We applied here the Friedman test [39] to check for differences in the classification accuracies across the different models and classifications (trinomial vs. binomial). A chi-squared value of 5 and a p-value of 0.025 suggested that there is a statistically significant difference in accuracies between the models and/or classifications since the p-value is less than the significance level of 0.05. Then we did a post-hoc Nemenyi test [40] to determine which specific pairs of models and classifications are significantly different. The result indicates that the accuracy difference between the model 1 and model 2 classifications is significant with a p-value of 0.025. This indicates that the binomial classification accuracy is significantly higher than the trinomial classification accuracy for the models tested.

Table 4
Result of Classification using different ML models with empirical data.

Models	Accuracy on Classification	
	Model 1	Model 2
Logistic Regression	0.752	0.875
Decision Tree	0.752	0.875
KNN	0.753	0.875
Linear Discriminant Analysis	0.752	0.875
Random Forest	0.743	0.869

3.3. Simulation results

The results in section 3.2 reveal that the test accuracy of the classifications are robust on the models but differ on the data and its distribution. From this motivation we perform simulation study by changing the distribution of the response variable to understand how the test accuracy differs in different models.

Table 5 presents the 10-fold cross-validation test accuracy for various ML models across different distributions of the response variable, both trinomial and binomial. When examining the row-wise comparison, it is evident that the accuracy fluctuates based on the distribution of the response variable. For instance, LR and LDA exhibit similar patterns, achieving their highest accuracy (0.799) for the 80-10-10 distribution (80 %, 10 % and 10 % of the data in each three categories of the response variable) in the trinomial case, indicating that these models perform well with highly imbalanced data. KNN and RF also follow a similar trend, though RF generally shows slightly lower accuracy than KNN across different distributions. Column-wise comparison highlights the impact of different response variable distributions on model performance. For the trinomial classification, as the distribution becomes more imbalanced (e.g., moving from 40 to 30-30 to 80-10-10), most models' accuracy improves, particularly LR and DT, suggesting that these models handle imbalanced classes effectively. Conversely, for binomial classification, the highest accuracy is consistently observed in the 90-10 distribution for most models, emphasizing that extreme imbalance favors these models. Overall, LR and LDA are notably robust across both trinomial and binomial classifications, showing high accuracy in imbalanced scenarios. In contrast, RF and KNN display variable performance, with KNN generally outperforming RF. This analysis underscores the importance of considering class distribution when selecting and tuning models for classification tasks.

We performed Friedman test and subsequent post-hoc Nemenyi test to provide valuable insights into the comparative performance of different classification models under various response variable distributions. The Friedman test yielded a chi-squared value of 49.119 with 10 degrees of freedom and a highly significant p-value of <0.001 . This indicates a statistically significant difference in the performance of the classification models across the different response variable distributions. The low p-value suggests that the null hypothesis, which states that all models perform equally, can be rejected.

Following the Friedman Test, the post-hoc Nemenyi Test was conducted to identify specific pairs of response variable distributions where significant differences in model performance exist. Table 6 presents the Nemenyi Test results, where comparisons between the response distributions 40-30-30(%) and 80-10-10(%), as well as 40-30-30(%) and 90-10(%), are found to be statistically significant with p-values of 0.003 and <0.001 , respectively. Additionally, significant differences are found between other pairs such as 40-40-20(%) and 90-10(%), 50-30-20(%) and 90-10(%), and so on, with p-values less than 0.05. These findings indicate that the performance of classification models varies considerably between certain response distributions, emphasizing the impact of the distribution of the response variable on model accuracy.

4. Discussion

The Bayesian multinomial mixed effect model results reveal substantial regional variability in the diabetic and prediabetic responses across the divisions of Bangladesh. The model incorporates random intercepts for eight divisions, allowing baseline odds of diabetes and prediabetes to vary regionally. The standard deviations of the intercepts for both diabetic and prediabetic responses with wide credible intervals underscore the importance of considering regional variability when designing public health interventions. These wide intervals reflect the uncertainty in these estimates but confirm substantial regional differences, suggesting that factors not directly measured in the study may influence diabetes prevalence in different divisions.

The prior predictive checks, illustrated in Fig. 1, indicate that our specified priors are reasonable for the model and data. The alignment of the predicted data with the observed data suggests that the priors are not overly restrictive and can generate a wide range of plausible data. This ensures that the model's assumptions are reasonable before fitting the actual data, a crucial step in Bayesian analysis to avoid unrealistic predictions and enhance the model's reliability. The analysis in Table 3 examines various demographic, socioeconomic, and health-related factors to understand their impact on diabetes and prediabetes. The significant associations identified, such as higher odds of diabetes among younger and middle-aged individuals, those with higher BMI, and those with hypertension, highlight key risk factors. Conversely, older individuals and rural residents have lower odds of diabetes. Wealthier individuals also show higher odds, emphasizing the role of socioeconomic status. For prediabetes, similar patterns are observed, with younger and middle-aged individuals and those from higher wealth quintiles showing increased odds. These results provide valuable

Table 5
Simulation results on test accuracy.

Models	Accuracy on Classification										
	Trinomial						Binomial				
	40 - 30 - 30 (%)	40 - 40 - 20 (%)	50 - 30 - 20 (%)	60 - 30 - 10 (%)	70 - 20 - 10 (%)	80 - 10 - 10 (%)	50 - 50 (%)	60 - 40 (%)	70 - 30 (%)	80 - 20 (%)	90 - 10 (%)
LR	0.389	0.39	0.503	0.608	0.705	0.799	0.499	0.608	0.705	0.799	0.900
KNN	0.363	0.364	0.394	0.493	0.588	0.729	0.500	0.532	0.606	0.737	0.881
RF	0.356	0.393	0.463	0.580	0.695	0.796	0.506	0.568	0.684	0.792	0.898
DT	0.394	0.406	0.507	0.608	0.705	0.799	0.493	0.608	0.705	0.799	0.900
LDA	0.389	0.389	0.507	0.608	0.705	0.799	0.499	0.608	0.705	0.799	0.900

Table 6
Result of post-hoc Nemenyi test.

	40-30-30 (%)	40-40-20 (%)	50-30-20 (%)	60-30-10 (%)	70-20-10 (%)	80-10-10 (%)	50-50 (%)	60-40 (%)	70-30 (%)	80-20 (%)
40-40-20 (%)	1.000	–	–	–	–	–	–	–	–	–
50-30-20 (%)	0.984	0.999	–	–	–	–	–	–	–	–
60-30-10 (%)	0.647	0.874	0.999	–	–	–	–	–	–	–
70-20-10 (%)	0.082	0.214	0.744	0.994	–	–	–	–	–	–
80-10-10 (%)	0.003	0.013	0.153	0.647	0.997	–	–	–	–	–
50-50(%)	0.984	0.999	1.000	0.999	0.744	0.153	–	–	–	–
60-40(%)	0.579	0.828	0.998	1.000	0.997	0.713	0.998	–	–	–
70-30(%)	0.082	0.214	0.744	0.994	1.000	0.997	0.744	0.997	–	–
80-20(%)	0.003	0.013	0.153	0.647	0.997	1.000	0.153	0.713	0.997	–
90-10(%)	0.000	0.001	0.018	0.192	0.852	1.000	0.018	0.237	0.852	1.000

insights into the relative likelihood of these conditions across different subgroups, informing targeted interventions.

The model diagnostics, including histograms and trace plots, confirm the convergence and reliability of the MCMC chains. The posterior distributions reveal the estimated effect sizes for each predictor, with the trace plots demonstrating well-mixed and converged chains. The high $\hat{R}$ values and ESS for most parameters indicate robust and reliable estimates. Parameters with lower ESS values could benefit from more iterations, but overall, the diagnostics suggest that the model's inferences are based on well-mixed chains with sufficient effective sample sizes. The $\hat{R}$ values below 1.05 and the high ESS ratios confirm the stability and reliability of the parameter estimates. The posterior predictive checks, depicted in Fig. 4, validate the model's fit by comparing the observed data to simulated data from the model's posterior predictive distribution. The close mirroring of the observed data by the simulated datasets indicates that the model effectively captures the underlying structure of the data. This alignment suggests a good model fit, with low uncertainty in the model's predictions, ensuring robust inferences and highlighting areas where the model performs well.

The 10-fold cross-validation results for various ML models, summarized in Table 4, demonstrate the accuracy of these models in classifying diabetic status using BDHS data. KNN slightly outperformed other models in trinomial classification, while LR, DT, and LDA performed similarly. RF showed slightly lower accuracy. For binomial classification, all models except RF achieved similar high accuracy. These results indicate that LR and LDA are notably robust across both classification tasks, while RF and KNN show more variable performance.

The simulation study, presented in Table 5, examines how different distributions of the response variable affect model performance. LR and LDA show high accuracy in highly imbalanced distributions, while KNN and random forest exhibit more variable performance. The study highlights the importance of considering class distribution when selecting and tuning models for classification tasks, as model accuracy can significantly vary with the distribution of the response variable. The Friedman test and subsequent Nemenyi test results, shown in Table 6, provide insights into the comparative performance of different classification models under various response variable distributions. Significant differences were identified between certain response distributions, emphasizing the impact of the distribution of the response variable on model accuracy. These findings highlight the necessity of accounting for response distribution when evaluating model performance and selecting appropriate classification techniques.

The results of this study provide a comprehensive analysis of the factors influencing diabetes and prediabetes in Bangladesh, but there are several avenues for further research and improvement. Firstly, this study excluded incomplete responses, which could introduce selection bias into the results. Future work could address this by employing imputation techniques such as multiple imputation by chained equation (MICE) [41] to estimate missing values and applying the Bayesian multinomial mixed-effects model to the full dataset, potentially enhancing the robustness and accuracy of the findings. Secondly, while this study utilized five ML models to predict diabetic status, there is scope for exploring a broader range of models, including advanced techniques like gradient boosting machines, neural networks, and ensemble methods. The application of automated ML [42] tools could also streamline model selection and hyperparameter tuning, thereby improving predictive performance. Lastly, the current analysis was limited by the availability of variables in the BDHS dataset. Incorporating additional clinical, lifestyle, and environmental variables could provide a more holistic understanding of diabetes risk factors and improve model accuracy. Future studies should aim to include a wider range of relevant variables to capture the multifaceted nature of diabetes, ultimately contributing to more effective public health strategies and interventions. This comprehensive approach will not only enhance predictive accuracy but also provide deeper insights into the underlying determinants of diabetes, thereby providing more targeted and effective public health initiatives.

The findings of this study have significant implications for both medical practice and society. The identification of key predictors such as BMI, age, hypertension, and socioeconomic factors in determining diabetes and prediabetes risk emphasizes the need for targeted interventions. From a medical perspective, the strong association of hypertension and obesity with diabetes underscores the importance of integrating regular blood pressure monitoring and weight management into diabetes prevention programs. Additionally, the variability in diabetes prevalence across regions highlights the necessity for region-specific public health strategies that address unique local challenges. On a societal level, the association of higher diabetes risk with wealthier individuals suggests that

urbanization and lifestyle changes may play a critical role, calling for educational campaigns on healthy dietary and exercise habits. These findings provide actionable insights that can guide healthcare providers and policymakers in designing comprehensive strategies to reduce the burden of diabetes and its complications, ultimately improving the quality of life and reducing healthcare costs.

5. Strengths and limitations

A major strength of this work is the use of the 2017–18 Bangladesh Demographic and Health Survey as the empirical study, one of the few nationally representative sources with contemporaneous biomarker data, ensuring that the findings are generalisable to the adult population of Bangladesh. The large sample size and two-stage stratified design delivered sufficient power to detect division-level heterogeneity, which we explicitly modelled through a Bayesian multinomial mixed effects framework with random intercepts for all eight divisions. By borrowing strength across clusters, this hierarchical specification captures unobserved contextual effects and yields more stable estimates than fixed effect approaches. Extensive model diagnostics, sensitivity analyses under alternative priors, and posterior predictive checks demonstrated excellent convergence and calibration, attesting to the robustness of the inferences. Also the parallel use of multiple supervised learning algorithms and a simulation study spanning a range of outcome prevalences confirmed that our substantive conclusions are not an artefact of any single analytic choice, underscoring the external validity of the results.

Several limitations should be recognized. First, the BDHS is cross-sectional; therefore, temporality cannot be established and causal claims regarding putative risk factors must be avoided. Second, although the survey collected rich socio demographic information, it did not record key clinical biomarkers (e.g., HbA1c, oral-glucose-tolerance tests, lipid parameters) or behavioural exposures such as diet and physical activity; residual confounding is therefore likely. Reliance on fasting plasma glucose more than 7 mmol/L, while aligned with WHO diagnostic criteria, may underestimate the true prevalence of diabetes because isolated post prandial hyperglycaemia and impaired glucose tolerance remain undetected in the absence of OGTT or HbA1c measurements. Third, the decision to perform complete case analysis, although methodologically transparent, introduces the possibility of selection bias if excluded respondents differ systematically from those retained. Finally, because the analysis is confined to the 2017–18 BDHS, any secular changes in diabetes prevalence occurring after that wave cannot be captured; however, our simulation study suggests that the core methodological conclusions are robust to plausible shifts in disease burden, supporting our plan to replicate the framework on the newly released 2022 BDHS in a dedicated follow-on paper.

6. Conclusions

This study provides an in-depth analysis of T2DM in Bangladesh by leveraging Bayesian multinomial mixed-effect models alongside various ML techniques to understand and predict diabetic and prediabetic status. Our approach underscores the importance of integrating these methodologies for effective diabetes prediction, offering valuable insights for public health strategies aimed at mitigating the impact of T2DM. The Bayesian multinomial mixed-effect model revealed substantial regional differences in diabetes and prediabetes responses across Bangladesh's divisions. The significant variability in baseline odds highlights the necessity of considering regional factors when designing public health interventions.

Our classification models, evaluated through 10-fold cross-validation, demonstrated that LR and LDA were notably robust across both trinomial and binomial classifications. The performance of KNN and RF models varied, with KNN generally outperforming RF. The simulation study further emphasized the importance of considering class distribution when selecting and tuning models, as model accuracy significantly varied with different response variable distributions. The Friedman test and post-hoc Nemenyi test provided insights into the comparative performance of different classification models under various response distributions, emphasizing the impact of response variable distribution on model accuracy.

In conclusion, this comprehensive approach of estimating parameters using a Bayesian model, applying ML for prediction, and conducting simulation studies to explore various scenarios, highlights the importance of integrating these methodologies for effective diabetes prediction. The significant regional variability and the influence of demographic, socioeconomic, and health-related factors underscore the complexity of managing diabetes. These findings support the development of targeted interventions and improved predictive models, ultimately enhancing public health outcomes and informing policy decisions. Future research should continue to explore the interplay of various predictors and refine these models to further improve the accuracy and reliability of diabetes predictions.

CRedit authorship contribution statement

Noor Muhammad Khan: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Md Mizanoor Rahman:** Writing – review & editing, Visualization, Validation, Resources, Formal analysis, Data curation, Conceptualization. **Mashfiqul Huq Chowdhury:** Writing – review & editing, Visualization, Validation, Supervision, Conceptualization.

Data availability statement

Data are available at <https://dhsprogram.com/>. Permission to use and analyze the data set was obtained by registering the study on the DHS website.

Ethics declaration

The study does not require ethical approval as the BDHS 2017–18 survey was approved by the local Ethics Committee of Bangladesh and the ICF Macro at Calverton, New York, USA. Permission to use and analyze the data set was obtained by registering for the study on the DHS website. Patients and the public were not involved in the design, conduct, reporting, or dissemination plans of this research.

Consent for publication

All authors have read and agreed to the published this manuscript.

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.heliyon.2026.e44510>.

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